

Scalability & Future Plan

Date	05 July 2026
Team ID	
Project Name	Personalized Networking Assistant
Maximum Marks	1 Mark

Current System Limitations Table:

This table captures the current limitations of the system and priority to address them.

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Local CPU Inference Latency	GPT-2 prompt generation can take up to 2 seconds under high load.	Medium
2	Single-user JSON Flat Storage	Lock contentions if multiple concurrent writes are issued.	High
3	Fixed Prompt Templates	Conversation prompts follow a static structure.	Low

Scalability Plan Table:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single local user Streamlit process.	Containerize with Docker, deploy behind Nginx reverse proxy.
2	Data Storage	Local JSON files (history/feedback).	Migrate to SQLite or PostgreSQL using SQLAlchemy ORM.
3	Performance	CPU PyTorch tensor operations.	Convert models to ONNX runtime or run on dedicated GPU.
4	Security	No client log encryption.	Implement OAuth2 JWT login authorization tokens.